

Multi-Criteria Decision Analysis of Autonomous Delivery Robot Prototypes Using TOPSIS

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0.1 Introduction

AutonomousShipment is a Leeds based start-up researching autonomous delivery robots. They have developed seven robot prototypes, each with distinct characteristics. The company is preparing for a trial, but due to resource constraints, can only select two prototypes to evaluate. This report will recommend two prototypes by addressing AutonomousShipment's two business plans: one derived from the management team's factor prioritisation based on maximising deliveries, and a second based on maximising robot sales, focusing on battery capacity, cost and reliability.

0.2 Data Understanding

The management team provided a dataset containing 7 observations of prototypes (Alpha, Bravo, Charlie, Delta, Echo, Foxtrot and Gamma) and 7 rows of criteria for robot capability outlined in Table 1.

Criteria	Description	Maximise/Minimise
Carrying Capacity	Carrying capacity per robot in litres.	Maximise
Battery Size	Battery capacity per robot in hours of operation.	Maximise
Speed	Average speed in km/h.	Maximise
Mobility	The subjective score out of five of ability to navigate with mobility.	Maximise
Aesthetic	The subjective score out of ten of aesthetic appearance.	Maximise
Cost per Unit	Cost per robot in GBP.	Minimise
Reliability	Estimated average time between breakdowns of each robot in days.	Maximise

Table 1: Description of each criteria and whether we aim to maximise or minimise each variable.

To better understand the data, Figure 1 compares the values for each criteria by prototype.

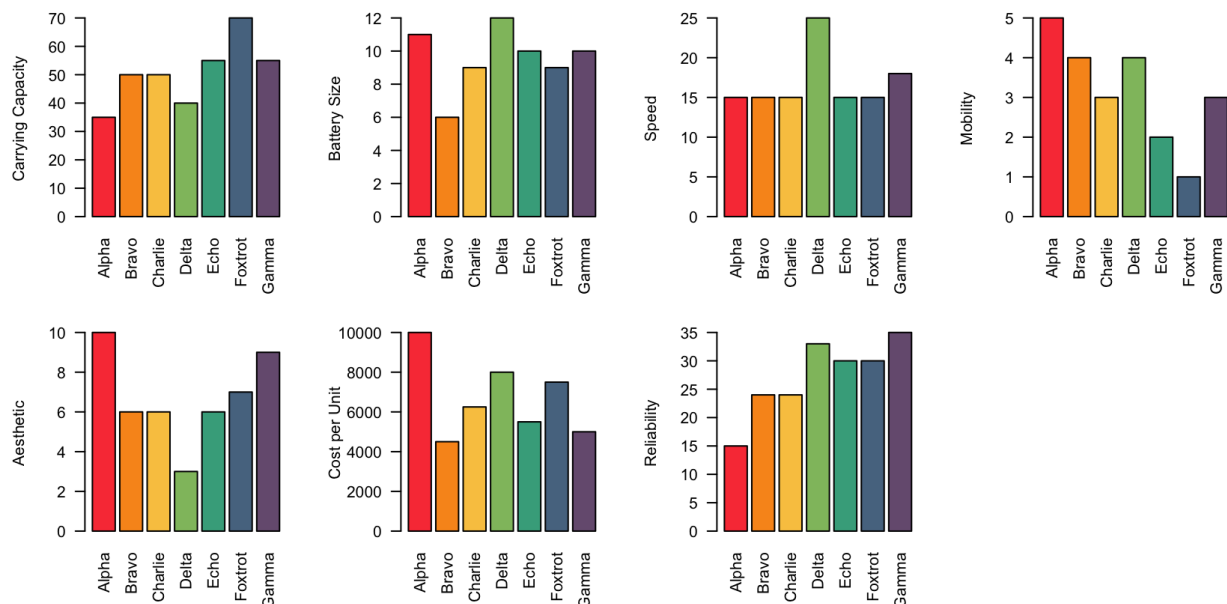


Figure 1: Bar plots of robot prototypes for each criterion.

Here we can see that each prototype has a varied characteristic profile, demonstrating that recommendations will likely vary when different criteria are prioritised.

A spreadsheet was also provided containing the management team's importance of each criterion for the first business plan, derived from stakeholder analysis (see Table 2).

Carrying Capacity	This is the most important criterion.
Battery Size	The battery size is more important than reliability, but not as much as mobility.
Speed	The speed is as important as the mobility of the vehicle.
Mobility	After careful deliberation, the mobility is rated 6 out of 10 in terms of importance. We believe that it is an important criterion, but not as important as some others, and that this would likely improve with better tech.
Aesthetic	This is the least important.
Cost Per Unit	The cost is the second most important criterion.
Reliability	This is slightly more important than the aesthetic.

Table 2: Table of the management team's importance of each criterion for the first business plan.

0.3 Multi-Criteria Decision Analysis (MCDA)

To answer the business questions in this report, the Technique for Order Preference by Similarity to Ideal Solutions (TOPSIS) method of MCDA was selected. Unlike additive models like the Weighted Sum Method (WSM), in which the utility of each alternative is simply aggregates, TOPSIS provides a more nuanced selection by both maximising distance to the negative ideal solution (NIS), and minimising distance to the positive ideal solution (PIS), decreasing the reliance on subjective clarity when weighting the criteria. TOPSIS also requires less data manipulation than methods like the Analytic Hierarchy Process (AHP), which rely on extensive pairwise comparisons. Transforming the robot prototype data in to this form would be a complex procedure and potentially result in a loss of information. The Viekriterijsko Kompromisno Rangiranje (VIKOR) method is similar to TOPSIS, but differs in its requirement for the value of a parameter, v , to be selected, dependent on whether the emphasis of analysis is to minimise distance to the PIS or maximise distance to the NIS. As this factor has not been specified in the AutonomousShipment brief, the use of TOPSIS eliminates this additional subjectivity.

0.3.1 Data Preparation

The `TOPSIS()` function in R requires three inputs: `performanceTable`, `criteriaWeights` and `criteriaMinMax`. The `performanceTable` requires each alternative per row and each criterion per column. As the original data was structured with criteria as rows and robot prototypes as columns, it was read in with the criteria as row names, then transposed using `t()` and converted back into a dataframe, producing the performance table in Table 3 with each robot prototype as a row and each criterion as a column.

	Carrying Capacity	Battery Size	Speed	Mobility	Aesthetic	Cost Per Unit	Reliability
Alpha	35	11	15	5	10	10000	15
Bravo	50	6	15	4	6	4500	24
Charlie	50	9	15	3	6	6250	24
Delta	40	12	25	4	3	8000	33
Echo	55	10	15	2	6	5500	30
Foxtrot	70	9	15	1	7	7500	30
Gamma	55	10	18	3	9	5000	35

Table 3: Performance table of robot prototypes.

For the `criteriaWeights` input, a vector of weights was produced, and named in accordance with the criterion, for each of the business plans. Tables 4 and 5 show the weights selected for the first and second business plans respectively.

Carrying Capacity	Battery Size	Speed	Mobility	Aesthetic	Cost Per Unit	Reliability
0.25	0.1	0.15	0.15	0.07	0.2	0.08

Table 4: Criteria weights for maximising number of large shipments.

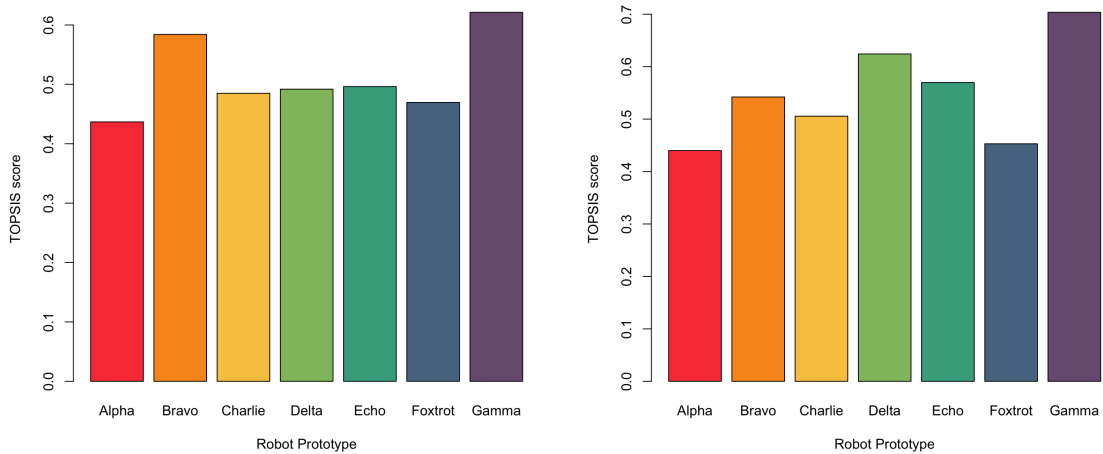
Carrying Capacity	Battery Size	Speed	Mobility	Aesthetic	Cost Per Unit	Reliability
0.1	0.2	0.12	0.12	0.06	0.2	0.2

Table 5: Criteria weights for prioritising selling the robot technology.

Finally, the `criteriaMinMax` input was constructed by specifying the preference elicitation as 'min' or 'max' in a vector, and naming it in accordance with the criteria.

0.3.2 TOPSIS Scores

Figure 2 visualises the prototype scores produced by the `TOPSIS()` function in R for both plans.



(a) Maximising number of large shipments scenario.

(b) Maximising robot technology sales scenario.

Figure 2: Barplots of TOPSIS scores for each robot prototype.

In both scenarios, the Gamma prototype achieves the highest TOPSIS score. Under the delivery-focused scenario, Gamma scores 0.621, followed closely by Bravo at 0.584. These two alternatives outperform the remaining prototypes by a notable margin. In the technology-focused TOPSIS scores, Gamma's lead is even more pronounced, with a score of 0.704, and there is a more uniform decline in performance across the top alternatives.

0.3.3 Sensitivity Analysis

To evaluate the robustness of the initial TOPSIS results, sensitivity analysis was conducted.

Local sensitivity was assessed by increasing and decreasing each weight by 0.06 in turn, with the corresponding change distributed evenly across the remaining criteria. For the maximising shipping

scenario, Gamma remained the top scoring prototype in all but one adjustment: when the weighting of mobility was increased, Bravo took over as the top recommendation. In the case of maximising robot technology sales, Gamma remained the top scoring prototype under all local weight perturbations.

The weights were then adjusted based on the subjectivity of their prioritisation. In the first scenario, the TOPSIS scores were calculated across alternative weighting schemes, including emphasis on the two highest priority criteria (Carrying Capacity and Cost Per Unit), and increasing the top four criteria (Carrying Capacity, Cost Per Unit, Speed and Mobility). Gamma remained the highest scoring prototype when only the two highest priority criteria were altered, however, when the weighting of the top four criteria were each increased by a critical threshold of 0.025, and the remaining three criteria decreased evenly, Bravo once again superseded Gamma. For the maximising sales model, alterations included cost-emphasis, reliability-emphasis and battery-emphasis scenarios, as well as weighting schemes with increased and decreased priority for these top three criteria. Under all of these tested perturbations, Gamma prevailed as the top prototype, further demonstrating the robustness of this recommendation.

To further assess the top prototypes, the weights from the two different scenarios were averaged. With these weights, Gamma retained its top scoring position (0.659), as expected, and Bravo was the second highest scoring prototype (0.572).

Finally, the TOPSIS scores were calculated with equal weights given to each criteria. Under these conditions, Gamma had a significant lead in TOPSIS score (0.657), followed by Alpha (0.539), then closely followed by Bravo (0.528).

0.4 Conclusion

This analysis identified Gamma as the top-performing prototype under both strategic scenarios. In the technology-focused plan, Gamma achieved a substantial lead in TOPSIS score and demonstrated strong robustness across all sensitivity tests, consistently remaining the highest-ranked alternative despite changes to subjective input weights. Thus, Gamma is recommended for trial under the second business plan provided by AutonomousShipment.

Gamma also scored highest in the shipments-focused scenario, with the weighting based on the criteria prioritisation provided by the management team, although Bravo's score was not far behind. Furthermore, this recommendation proved less robust: Bravo overtook Gamma when mobility was prioritised in local sensitivity analysis, as well as when the weighting of the top four criteria was increased. Given the requirement to recommend two distinct prototypes, Bravo is selected for trial under the first business plan. This choice is further supported by Bravo's consistent performance across scenarios, including placing second when averaging the weights from both plans, and a top-three ranking under equal criteria weighting, indicating its strength as a well-rounded alternative.